Rubric for Final Project Presentation

For your final presentation, you will have a 25 minute presentation either on slides or blackboard. You can also use a combo: eg. mostly use slides, but write some things on the board (this is a good way to remind the audience of key terms or definitions), or mostly use the board, but use slides just to show a figure.

There are two new (intertwined) emphases for this presentation: SCOPE and STORY.

1) **Scope:** Since your paper is long and complex, you will have to decide what you want to highlight in your short presentation. What will be accessible to your classmates and fun for them to learn?
2) **Story:** A good presentation should have an arc that motivates (or builds tension towards) the main result. Why is what you are telling us interesting, important, or surprising? I recommend reading this short piece on storytelling in math presentations. (this was also useful.)

You are welcome but not required to schedule a practice presentation with me or Matt.

AI Policy.
The purpose of this presentation is for you to learn how to:

- Reduce a complex body of work into a compelling and accessible story.
- Design a presentation that focuses on your main story.
- Use examples or visuals to demonstrate challenging mathematical concepts.

Where you may want to use AI:
AI can be a helpful assistant in helping you understand the text and creating visuals, but you should be heavily involved in the process. For example, to generate an example, you might ask AI questions like “can you give me an example of a hypothesis class where the SSP lemma is not tight”, and then converse to refine the example. Or, you might say “Can you create a figure showing why this particular inequality holds”.

Where you may not use AI:
You may NOT use AI to make the slides or dedicate the presentation. You may copy-paste (with a citation) a figure generated using AI into your slide, or ask AI for formatting help which you execute yourself. You should interface with the software yourself so that you can learn what tools are available. I’ve found that content created by AI is much harder to present because it does not follow your own intuitions for the flow of how to convey ideas.

	Exceeds (100%)	Meets (80%)	Below (40%)
Preparation (3/10)	The presentation has an arc that motivates the main contribution. The scope of the presentation was appropriate for the class. It is clear that the presenter understood the texts' main ideas and these main ideas were clearly conveyed using titles, emphasis, and repetition.	The presenter understood the main ideas but they were initially hard for the listener to distill.	Hard to follow main ideas of presentation
Organization (1.5/10)	Outline, summary, and additional verbal cues marked transitions to keep the listener on track. Each slide/board had a clear title conveying its purpose. *For slides*: Key information was recalled (copied onto new slide) appropriately. *For board:* Writing on the board was organized and legible.	Used some of the "exceeds" organizational tools, but sometimes hard for the listener to understand where we were or where we were going.	Lack of organization made talk hard to follow.
Math Correctness and Completeness (1/10)	Stated definitions and theorems fully, fully defined notation, and proofs were correct. Any omitted steps or lemmas were clearly pointed out.	Some missing details or errors, but were able to fill these in when asked.	Missing information or errors made the talk hard to follow.
Easy to follow math including visual (2.5/10)	Clearly explained proof steps and motivations. Used figure(s) or examples with a clear purpose. *For board presentations,* you only need examples. Included connections to the course content when possible.	Clearly explained proof steps and why they were valid, and used at least 1 figure/example.	Copied proof from notes with no additional explanations or figures.
Pacing + Engaged with Audience (2/10)	Made eye contact, is aware if the audience is following, paused for questions and answered clearly. *For slides:* Used well-paced slides where content was revealed at a digestible pace (eg. by using animations)	Most of "exceeds" with some engagement or pacing challenges.	Did not engage with the audience, or the pacing of slides made the presentation too hard to follow.